**Statistics for Data Science - 2**

**Week 8 practice Assignment**

**Statistics from samples and Limit theorems**

1. If $X, Y \sim$ i.i.d. Normal$(0, 4)$, what will be the variance of $\frac{X}{Y}$?

   (a) 4

   (b) 2

   (c) 1

   (d) Undefined

   **Solution:**
   We know that if $X, Y \sim$ i.i.d. Normal$(0, \sigma^2)$, $\frac{X}{Y} \sim$ Cauchy$(0, 1)$ and variance of Cauchy distribution is undefined.
   Therefore, option(d) is correct.

---

2. A study shows that the average daily sleeping hours of teenagers is ten hours with a standard deviation of two hours. If a sample of 100 teenagers is collected, what will be the probability that the mean of the sleeping hours of these 100 teenagers is at least 0.4 hours away from the population mean? Assume that each observation in the sample is independent. Assume that $F_Z$ denotes the CDF of standard normal distribution.

   (a) $1 + F_Z(-2) - F_Z(2)$

   (b) $1 - F_Z(-2) + F_Z(2)$

   (c) $F_Z(2) - F_Z(-2)$

   (d) $F_Z(2)$

   **Solution:**
   let $X$ denote the average daily sleeping hours of teenagers.
   Given: standard deviation of $X$, $\sigma = 2$
   Sample size, $n = 100$

   To find: $P(|\overline{X} - \mu| \geq 0.4)$ where $\overline{X}$ and $\mu$ are sample mean and population mean, respectively.

   Let $S = X_1 + X_2 + \ldots X_{100}$ where $X_i$ denotes the $i$th sample.
   By CLT, we know that $\frac{S - n\mu}{\sigma\sqrt{n}} \sim$ Normal$(0, 1) \Rightarrow \frac{S - 100\mu}{20} \sim Z$ (Standard Normal)

Now,

$$
\begin{aligned}
P(|\overline{X}-\mu| \geq 0.4) &= P(\left|\frac{S}{n}-\mu\right| \geq 0.4) \\
&= P(\left|\frac{S-n\mu}{n}\right| \geq 0.4) \\
&= P(\left|\frac{S-n\mu}{\sigma\sqrt{n}}\right| \geq \frac{0.4\sqrt{n}}{\sigma}) \\
&= P(|Z| \geq 2) \\
&= P(Z \geq 2) + P(Z \leq -2) \\
&= 1 - P(Z \leq 2) + P(Z \leq -2) \\
&= 1 - F_Z(2) + F_Z(-2)
\end{aligned}
$$

---

3. What is the fourth moment of the Normal$(0, 4)$ distribution?
**Solution:**
$M_X(\lambda) = E[e^{\lambda X}] = E[1 + \lambda X + \frac{\lambda^2 X^2}{2!} + \frac{\lambda^3 X^3}{3!} + ...]$

$$= 1 + \lambda E[X] + \frac{\lambda^2 E[X^2]}{2!} + \frac{\lambda^3 E[X^3]}{3!} + ...$$

In the moment generating function, coefficient of $\lambda$ will give first moment ($E[X]$), coefficient of $\frac{\lambda^2}{2!}$ will give the second moment ($E[X^2]$) and similarly, coefficient of $\frac{\lambda^k}{k!}$ will give the $k$th moment ($E[X^k]$).

Moment generating function of Normal$(0, \sigma^2)$ is given by $e^{\lambda^2\sigma^2/2}$.
Let $N \sim$ Normal$(0, 2^2)$

$$
\begin{aligned}
M_N(\lambda) &= e^{\lambda^2 2^2/2} \\
&= 1 + \frac{\lambda^2 2^2}{2} + \frac{\lambda^4 2^4}{2!(4)} + \dots \\
&= 1 + \frac{\lambda^2 2^2}{2} + 48\frac{\lambda^2}{4!} + \dots
\end{aligned}
$$

Therefore, 4th moment of Normal$(0, 2^2)$ = coefficient of $\frac{\lambda^4}{4!} = 48$

---

4. Let $X \sim$ Gamma$(2, \frac{1}{2})$ and $Y \sim$ Gamma$(5, \frac{1}{2})$ be two independent random variables. What will be the expected value of $\frac{X}{X+Y}$? Write your answer correct to two decimal

places.

**Solution:**

We know that if $X \sim \text{Gamma}(\alpha, k)$ and $Y \sim \text{Gamma}(\beta, k)$ be two independent random variables, then $\dfrac{X}{X+Y} \sim \text{Beta}(\alpha, \beta)$.

Given that $X \sim \text{Gamma}(2, \frac{1}{2})$ and $Y \sim \text{Gamma}(5, \frac{1}{2})$ are two independent random variables. It implies that

$$\frac{X}{X+Y} \sim \text{Beta}(2, 5)$$

Therefore, $E\left[\dfrac{X}{X+Y}\right] = \dfrac{2}{2+5} = 0.28$

---

5. A company sells eggs whose weights are normally distributed with a mean of 70g and a standard deviation of 2g. Suppose that these eggs are sold in packages that each contain four eggs. Assume that the weight of each egg is independent. What is the probability that the mean weight of the four eggs in a package is greater than 68.5g? Write your answer correct to two decimal places.
(Hint: Use the fact that linear combination of normal distributions is again a normal distribution. $F_Z(-1.5) = 0.066$)

**Solution:**

Let $X$ denote the weight of an egg.
Given that $E[X] = \mu = 70$
$\text{SD}(X) = \sigma = 2$
$X \sim \text{Normal}(70, 2^2)$ Let $X_1, X_2, X_3$ and $X_4$ denote the weights of four eggs in a package.

Suppose that
$\overline{X} = \dfrac{X_1 + X_2 + X_3 + X_4}{4}$

To find: $P(\overline{X} > 68.5)$

We know that linear combination of independent Normal distribution is again a Normal distribution.
It implies that $\overline{X}$ is a Normal distribution.

$E[\overline{X}] = \mu = 70$ and
$\text{Var}(\overline{X}) = \dfrac{\sigma^2}{n} = \dfrac{4}{4} = 1$

It implies that $\overline{X} \sim \text{Normal}(70, 1) \Rightarrow \overline{X} - 70 \sim \text{Normal}(0, 1)$

Now,

$$\begin{aligned} P(\overline{X} > 68.5) &= P(\overline{X} - 70 > -1.5) \\ &= P(Z > -1.5) \\ &= 1 - F_Z(-1.5) \\ &= 1 - 0.066 = 0.93 \end{aligned}$$

---

6. Let $X_1, X_2, X_3, \ldots X_n$ be i.i.d. Poisson(4). What should be the value of $n$ such that $P(3.8 \leq \overline{X} \leq 4.2) \geq 0.95$? [2 marks]
(Hint: Use $F_Z(1.96) = 0.975$)

   1. at least 200
   2. at least 385
   3. at least 450
   4. at least 585

**Solution:**
Given that $X_1, X_2, X_3, \ldots X_n \sim$ i.i.d. Poisson(4)

Mean of the distribution $= \mu = 4$
Variance of the distribution $= \sigma^2 = 4$
Let $S = X_1 + X_2 + \ldots + X_n$ and
$\overline{X} = \dfrac{X_1 + X_2 + \ldots + X_n}{n}$

To find: value of $n$ such that $P(3.8 \leq \overline{X} \leq 4.2) \geq 0.95$
By CLT, we know that
$\dfrac{S - n\mu}{\sqrt{n}\sigma} \sim \text{Normal}(0, 1)$
$\Rightarrow \dfrac{S - 4n}{2\sqrt{n}} \sim \text{Normal}(0, 1)$ ...(1)

$$
\begin{aligned}
&P(3.8 \leq \overline{X} \leq 4.2) \geq 0.95 \\
\Rightarrow &P(3.8 \leq \frac{S}{n} \leq 4.2) \geq 0.95 \\
\Rightarrow &P(-0.2 \leq \frac{S}{n} - 4 \leq 0.2) \geq 0.95 \\
\Rightarrow &P(-0.2 \leq \frac{S-4n}{n} \leq 0.2) \geq 0.95 \\
\Rightarrow &P(-0.1 \leq \frac{S-4n}{2n} \leq 0.1) \geq 0.95 \\
\Rightarrow &P(-0.1\sqrt{n} \leq \frac{S-2n}{2\sqrt{n}} \leq 0.1\sqrt{n}) \geq 0.95 \\
\Rightarrow &F_Z(0.1\sqrt{n}) - F_Z(-0.1\sqrt{n}) \geq 0.95 \\
\Rightarrow &F_Z(0.1\sqrt{n}) - (1 - F_Z(0.1\sqrt{n})) \geq 0.95 \\
\Rightarrow &2F_Z(0.1\sqrt{n}) - 1 \geq 0.95 \\
\Rightarrow &F_z(0.1\sqrt{n}) \geq 0.975 \\
\Rightarrow &0.1\sqrt{n} \geq 1.96 \\
\Rightarrow &n \geq 384.16
\end{aligned}
$$

---

7. Let the moment generating function of a random variable $X$ be given by

$$M_X(\lambda) = \left(\frac{1}{8}\right) e^{-4\lambda} + \left(\frac{1}{6}\right) e^{-2\lambda} + \left(\frac{1}{6}\right) e^{2\lambda} + \left(\frac{1}{8}\right) e^{4\lambda} + \left(\frac{5}{12}\right)$$

Find the distribution of $X$. [1 mark]

| $X$ | $-4$ | $-2$ | $0$ | $2$ | $4$ |
|---|---|---|---|---|---|
| $P(X=x)$ | $\frac{1}{8}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{8}$ | $\frac{5}{12}$ |

1.

| $X$ | $-4$ | $-2$ | $0$ | $2$ | $4$ |
|---|---|---|---|---|---|
| $P(X=x)$ | $\frac{5}{12}$ | $\frac{1}{8}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{8}$ |

2.

| $X$ | $-4$ | $-2$ | $0$ | $2$ | $4$ |
|---|---|---|---|---|---|
| $P(X=x)$ | $\frac{1}{8}$ | $\frac{1}{6}$ | $\frac{5}{12}$ | $\frac{1}{6}$ | $\frac{1}{8}$ |

3.

| $X$ | $-4$ | $-2$ | $0$ | $2$ | $4$ |
|---|---|---|---|---|---|
| $P(X=x)$ | $\frac{1}{8}$ | $\frac{1}{6}$ | $\frac{5}{12}$ | $\frac{1}{8}$ | $\frac{1}{6}$ |

4.

**Solution:**
The MGF of a discrete random variable $X$ with the PMF $f_X(x) = P(X = x)$, $x \in T_X$ is given by

$$\begin{aligned} M_X(\lambda) &= E[e^{\lambda X}] \\ &= \sum_{x \in T_X} P(X = x).e^{\lambda x} \end{aligned}$$

Now, MGF of a random variable $X$ be given by

$$M_X(\lambda) = \left(\frac{1}{8}\right) e^{-4\lambda} + \left(\frac{1}{6}\right) e^{-2\lambda} + \left(\frac{1}{6}\right) e^{2\lambda} + \left(\frac{1}{8}\right) e^{4\lambda} + \left(\frac{5}{12}\right)$$

Therefore, distribution of $X$ is given by

| $X$ | $-4$ | $-2$ | $0$ | $2$ | $4$ |
|---|---|---|---|---|---|
| $P(X=x)$ | $\frac{1}{8}$ | $\frac{1}{6}$ | $\frac{5}{12}$ | $\frac{1}{6}$ | $\frac{1}{8}$ |

---

8. A fair die is rolled 3600 times. Use CLT to compute the probability that six appears at most 630 times. Enter the answer correct to two decimal places.
(Hint: Use $F_Z(1.341) = 0.91$)
**Solution:**
Define a random variable $X$ such that

$$X = \begin{cases} 1 & \text{if six appears on rolling a fair die} \\ 0 & \text{otherwise} \end{cases}$$

Therefore, $E[X] = \mu = \dfrac{1}{6}$ and
$\text{Var}(X) = \sigma^2 = \dfrac{1}{6}.\dfrac{5}{6} = \dfrac{5}{36}$

Let $X_1, X_2, \ldots, X_{3600}$ be outcomes on rolling the fair die 3600 times.
Notice that $X_1 + X_2 + \ldots + X_{3600}$ will denote the number of times six appears in 3600 rolls.

Let $S = X_1 + X_2 + \ldots + X_{3600}$

To find: $P(S \leq 630)$

By CLT, we know that

$$\frac{S - 3600\mu}{\sigma\sqrt{n}} \sim \text{Normal}(0, 1)$$

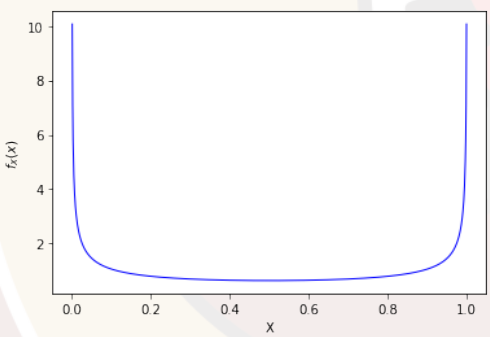
$$\Rightarrow \frac{S - 600}{10\sqrt{5}} \sim \text{Normal}(0, 1)$$

Now,

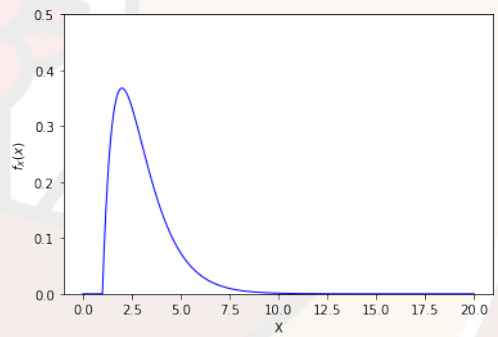
$$\begin{aligned} P(S \leq 630) &= P(S - 600 \leq 30) \\ &= P(\frac{S - 600}{10\sqrt{5}} \leq \frac{30}{10\sqrt{5}}) \\ &= P(Z \leq 1.34) \\ &= 0.91 \end{aligned}$$

---

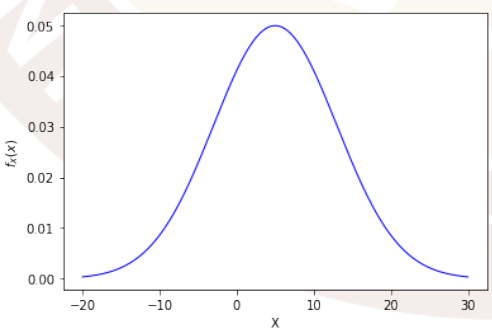
9. Consider the following PDF curves and match them with the correct distribution. [1 mark]



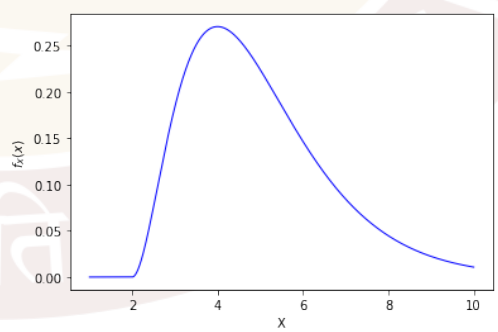

Graph 1




Graph 2




Graph 3




Graph 4

(a) Graph 1 $\rightarrow$ Gamma, Graph 2 $\rightarrow$ Normal, Graph 3 $\rightarrow$ Gamma, Graph 4 $\rightarrow$ Beta.

(b) Graph 1 $\rightarrow$ Beta, Graph 2 $\rightarrow$ Gamma, Graph 3 $\rightarrow$ Normal, Graph 4 $\rightarrow$ Gamma.

(c) Graph 1 $\rightarrow$ Beta, Graph 2 $\rightarrow$ Normal, Graph 3 $\rightarrow$ Normal, Graph 4 $\rightarrow$ Gamma.

(d) Graph 1 $\rightarrow$ Gamma, Graph 2 $\rightarrow$ Normal, Graph 3 $\rightarrow$ Normal, Graph 4 $\rightarrow$ Beta.

**Solution:**
Graph 1: Range of the distribution is [0, 1] and shape of the graph resembles to the Beta distribution.

Graph 2: PDF curve is not symmetric about mean and shape of the graph resembles to the Gamma distribution.

Graph 3: PDF curve is symmetric about mean and shape of the graph resembles to the Normal distribution.

Graph 4: PDF curve is not symmetric about mean and shape of the graph resembles to the Gamma distribution.
Therefore, Graph 1 $\rightarrow$ Beta, Graph 2 $\rightarrow$ Gamma, Graph 3 $\rightarrow$ Normal, Graph 4 $\rightarrow$ Gamma.

---

10. Let $X_1, X_2$ and $X_3 \sim$ i.i.d. $X$ where $X$ has the following probability mass function:

| $x$ | -1 | 2 |
|---|---|---|
| $f_X(x)$ | $\frac{2}{3}$ | $\frac{1}{3}$ |

Table 7.1.P: PMF of $X$

Find the distribution of $Y = X_1 + X_2 + X_3$. [1 mark]

(a)

| $Y$ | -3 | 0 | 3 | 6 |
|---|---|---|---|---|
| $P(Y=y)$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{3}$ | $\frac{1}{3}$ |

(b)

| $Y$ | -3 | 0 | 3 | 6 |
|---|---|---|---|---|
| $P(Y=y)$ | $\frac{8}{27}$ | $\frac{4}{9}$ | $\frac{2}{9}$ | $\frac{1}{27}$ |

(c)

| $Y$ | -3 | 0 | 3 | 6 |
|---|---|---|---|---|
| $P(Y=y)$ | $\frac{8}{27}$ | $\frac{1}{27}$ | $\frac{4}{9}$ | $\frac{2}{9}$ |

(d)

| $Y$ | -3 | 0 | 3 | 6 |
|---|---|---|---|---|
| $P(Y=y)$ | $\frac{2}{9}$ | $\frac{8}{27}$ | $\frac{1}{27}$ | $\frac{4}{9}$ |

**Solution:**
The PMF of $X$ is given by

| $x$ | -1 | 2 |
|---|---|---|
| $f_X(x)$ | $\frac{2}{3}$ | $\frac{1}{3}$ |

Given that $Y = X_1 + X_2 + X_3$ where $X_1, X_2$ and $X_3 \sim$ i.i.d. $X$.
To find: Distribution of $Y$.

We will find the distribution of $X$ by finding the MGF of $Y$.

$$\begin{aligned}
M_Y(\lambda) &= E[e^{\lambda Y}] \\
&= E[e^{\lambda(X_1+X_2+X_3)}] \\
&= E[e^{\lambda X_1} e^{\lambda X_2} e^{\lambda X_3}] \\
&= E[e^{\lambda X_1}]E[e^{\lambda X_2}]E[e^{\lambda X_3}] \qquad \text{(Since, } X_1, X_2 \text{ and } X_3 \text{ are independent)} \\
&= E[e^{\lambda X}]E[e^{\lambda X}]E[e^{\lambda X}] \qquad \text{(Since, } X_1, X_2 \text{ and } X_3 \sim \text{i.i.d. } X) \\
&= [M_X(\lambda)]^3 \qquad ...(1)
\end{aligned}$$

Now,

$$\begin{aligned}
M_X(\lambda) &= E[e^{\lambda X}] \\
&= e^{-1\lambda}.P(X = -1) + e^{2\lambda}.P(X = 2) \\
&= \frac{2e^{-\lambda}}{3} + \frac{e^{2\lambda}}{3} \qquad ...(2)
\end{aligned}$$

From equation (1) and (2), we have

$$\begin{aligned}
M_Y(\lambda) &= \left(\frac{2e^{-\lambda}}{3} + \frac{e^{2\lambda}}{3}\right)^3 \\
&= \frac{1}{27}(2e^{-\lambda} + e^{2\lambda})^3 \\
&= \frac{1}{27}(8e^{-3\lambda} + e^{6\lambda} + 12e^{-2\lambda}e^{2\lambda} + 6e^{-\lambda}e^{4\lambda}) \qquad \text{(since, } (a+b)^3 = a^3 + b^3 + 3a^2b + 3ab^2) \\
&= \frac{8}{27}e^{-3\lambda} + \frac{1}{27}e^{6\lambda} + \frac{4}{9} + \frac{2}{9}e^{3\lambda}
\end{aligned}$$

Therefore, distribution of $Y$ is given by

| $Y$ | -3 | 0 | 3 | 6 |
|---|---|---|---|---|
| $P(Y = y)$ | $\frac{8}{27}$ | $\frac{4}{9}$ | $\frac{2}{9}$ | $\frac{1}{27}$ |